Class 9 Notes

Quick revision: Maths, English, Science

Maths

- **Number systems:** Real numbers = rational + irrational. Rational: p/q form, decimal is terminating or repeating. Irrational: non-terminating and non-repeating (√2, π). Rationalise the denominator: $1/(\sqrt{a} + \sqrt{b}) = (\sqrt{a} - \sqrt{b})/(a - b)$.
- **Polynomials:** Degree is the highest power. Remainder theorem: p(x) divided by (x - a) leaves remainder p(a). Factor theorem: (x - a) is a factor if p(a) = 0.
- **Identities:** $(x + y)^2 = x^2 + 2xy + y^2$. $x^3 + y^3 = (x + y)(x^2 - xy + y^2)$. $x^3 - y^3 = (x - y)(x^2 + xy + y^2)$. $(x + y + z)^2 = x^2 + y^2 + z^2 + 2xy + 2yz + 2zx$.
- **Coordinate geometry:** Quadrants: I (+,+), II (-,+), III (-,-), IV (+,-). A point (x, y): x is abscissa, y is ordinate.
- **Linear equations:** ax + by + c = 0 represents a straight line. Infinitely many solutions lie on that line.
- **Lines and angles:** Linear pair = 180°. Vertically opposite angles are equal. Parallel lines cut by a transversal: corresponding, alternate and co-interior angles. Angle sum of a triangle = 180°.
- **Triangles:** Congruence: SAS, ASA, AAS, SSS, RHS. Angles opposite equal sides are equal. The larger side is opposite the larger angle.
- **Quadrilaterals:** Mid-point theorem: the line joining the mid-points of two sides is parallel to the third side and half of it. Diagonals of a parallelogram bisect each other.
- **Circles:** Angle at the centre is double the angle on the circle. Angles in the same segment are equal. Cyclic quadrilateral: opposite angles add to 180°.
- **Heron's formula:** Area = $\sqrt{[s(s - a)(s - b)(s - c)]}$, where $s = (a + b + c)/2$.
- **Surface area and volume:** Sphere: $SA = 4\pi r^2$, $V = (4/3)\pi r^3$. Cone: $CSA = \pi rl$, $l^2 = r^2 + h^2$, $V = (1/3)\pi r^2 h$. Hemisphere: $CSA = 2\pi r^2$, $TSA = 3\pi r^2$, $V = (2/3)\pi r^3$.
- **Statistics:** Mean, median, mode. Class intervals, frequency table, bar graph, histogram, frequency polygon.

English

- **Tenses:** Use the right tense in narration and in 'if' sentences. Avoid mixing past and present.
- **Subject-verb agreement:** Each, either, neither, everyone take singular verbs. 'A number of' takes a plural verb; 'the number of' takes a singular verb.
- **Reported speech:** Statements, questions, commands and requests. Change pronouns, tense and time words (now → then, here → there).
- **Active and passive:** Object becomes subject. The passive verb uses 'be' + past participle. Commands and questions follow special patterns.
- **Determiners and modals:** Use few, a few, little, a little, the, no article. Modals show ability, advice, possibility, obligation.
- **Gap filling and editing:** Read the sentence fully. Check the tense, article, preposition and agreement. In editing, find and correct the error.
- **Writing:** Formal letter, article, diary entry, story and descriptive paragraph. Keep to the word limit, use a clear opening, middle and ending.

- **Reading skills:** Skim for the main idea, scan for facts. Answer in your own words. Use synonyms for difficult words.

Science

- **Matter:** Solid, liquid, gas. Melting point of ice 273 K (0°C), boiling point of water 373 K (100°C). K = °C + 273. Evaporation causes cooling.
- **Pure substances and mixtures:** Element and compound are pure. Mixtures are homogeneous (solution) or heterogeneous. Solution, suspension and colloid differ by particle size. Separation: filtration, distillation, chromatography, evaporation.
- **Atoms and molecules:** Law of conservation of mass; law of constant proportions. 1 mole = 6.022×10^{23} particles. Molar mass is in grams.
- **Structure of the atom:** Thomson: plum pudding. Rutherford: nucleus. Bohr: orbits. Chadwick: neutron. Z = number of protons; A = protons + neutrons. Electron distribution 2, 8, 8. Isotopes have the same Z but different A.
- **Cell:** Prokaryotic (no nucleus) and eukaryotic cells. Mitochondria: powerhouse. Ribosomes: proteins. Lysosomes: waste. Chloroplast: photosynthesis (plants only).
- **Tissues:** Plant: meristematic (growth), permanent (simple and complex). Animal: epithelial, connective, muscular, nervous.
- **Motion:** Speed = distance/time, velocity = displacement/time. Acceleration $a = (v - u)/t$. Equations: $v = u + at$, $s = ut + (1/2)at^2$, $v^2 = u^2 + 2as$.
- **Laws of motion:** First: inertia. Second: $F = ma$. Third: every action has an equal and opposite reaction. Momentum $p = mv$. Momentum is conserved in collisions.
- **Gravitation:** $F = G m_1 m_2/r^2$. $G = 6.67 \times 10^{-11}$ N m²/kg². $g = 9.8$ m/s². Weight = mg. Buoyant force acts upward; Archimedes' principle.
- **Work and energy:** $W = F \times s$ (joule). $KE = (1/2)mv^2$. $PE = mgh$. Power = work/time (watt). Energy is conserved.
- **Sound:** Longitudinal wave (compressions and rarefactions). $v = f \times \lambda$. Audible range 20 Hz to 20,000 Hz. Echo needs at least 17 m distance. Ultrasound is above 20,000 Hz.
- **Natural resources:** Water cycle, nitrogen cycle, carbon cycle. Greenhouse effect warms Earth. Ozone layer protects from UV rays.